

ACTIVITY 12 — LEARNER WORKSHEET**Network Diagram, PERT and the Critical Path**

TYPE	DURATION	MAPS TO	TOPIC
Workshop	90 minutes	LO2	3 - Plan the Project

What you are doing

Contoso Learning Pte Ltd is a training provider with 14,000 active learners. Its course registration portal is 9 years old. Registration takes a median of 11 minutes and 34% of attempts are abandoned. Learner communications consume 24 staff-hours per week across 3 administrators. 62% of learner traffic is mobile, against a portal that is not responsive. Legacy consent screens do not meet current PDPA requirements. The project modernises course registration and learner communications.

What you will produce: activity list, network diagram, critical path — submitted to artifacts/.

What you need before you start

- Lab 11 WBS and work packages
- Lab 03 lessons learned (compliance review must precede UAT)

Your steps

1. Derive the activity list from the WBS. Lab 11 produced deliverables. A schedule needs activities - the verbs that produce those nouns. This is the conversion the WBS deliberately did not do.
2. Understand the dependencies you have just declared
3. Draw the network diagram. Create artifacts/12-network-diagram.md. This is an activity-on-node diagram: activities are boxes, arrows are dependencies.
4. Compute the forward and backward pass. Do this by hand before reading the table. The method:
5. Apply three-point estimating and compute confidence ranges. Single-point estimates hide uncertainty. Three-point estimating exposes it.
6. Compress the schedule. Crash cost table. All four activities below are on the critical path.
7. Apply the Lab 03 lesson: compliance before UAT. The Lab 03 lessons-learned register recorded that on the previous release, compliance review was run concurrently with user acceptance testing to save time, and the findings forced rework of screens that had already been accepted by users. The lesson: compliance review must complete before UAT begins.
8. Build the schedule baseline. Create artifacts/12-schedule-baseline.md.
9. Answer the exam-style scenarios

Your working space

Complete every field. Keep this worksheet — it is your reference for the rest of the course.

Step 1 — Derive the activity list from the WBS**Step 2 — Understand the dependencies you have just declared**

Step 3 — Draw the network diagram

Step 4 — Compute the forward and backward pass

Step 5 — Apply three-point estimating and compute confidence ranges

Step 6 — Compress the schedule

Step 7 — Apply the Lab 03 lesson: compliance before UAT

Step 8 — Build the schedule baseline

Step 9 — Answer the exam-style scenarios

Deliverable checklist

- 12-activity-list.md - at least 14 activities with IDs, WBS traces, durations, predecessors and owners, each expressed as a verb phrase.
- 12-network-diagram.md - the activity-on-node diagram with all paths identified and their lengths computed.

- 12-critical-path.md - the full forward and backward pass table with ES, EF, LS, LF, total float, free float and the critical flag for every activity; the critical path named with its duration; near-critical paths identified with their float.
- The three-point estimating table with both triangular and beta/PERT computed, standard deviations and variances, and at least one skewed example showing the two formulas diverging.
- The project-level standard deviation computed from critical path variances only, with the 1, 2 and 3 sigma confidence ranges stated in words.
- The crash analysis with cost per day saved for at least four activities and the cheapest three-day compression selected and justified.
- The compliance-before-UAT analysis with the parallel-running duration computed and the realistic rework outcome quantified.
- 12-schedule-baseline.md - all components listed in Step 8.
- Written answers to the three scenarios.